

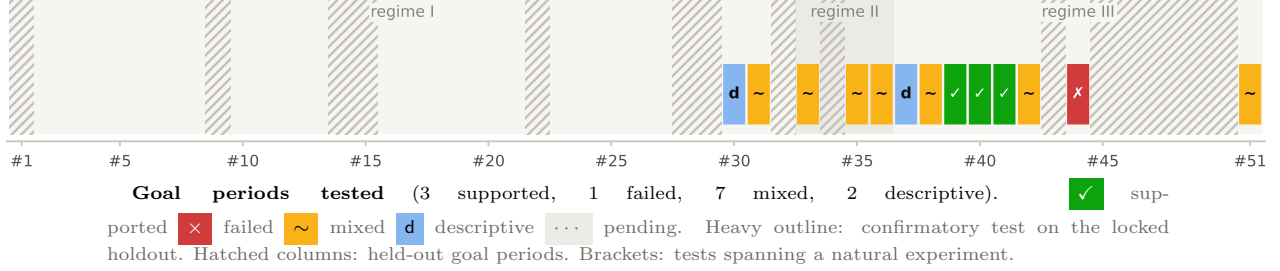
H94: Work allocation as a max-entropy equilibrium

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 40% Faithfulness 2.5/5 Usefulness 2/5

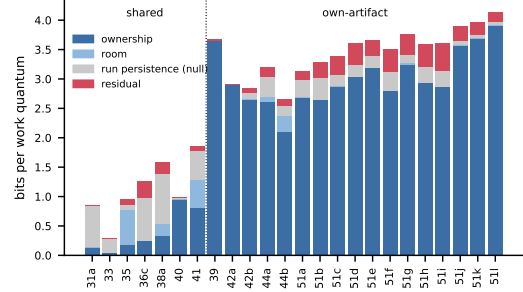
Bottom line. Work episodes are allocated at maximum entropy given agent activity, repo sizes, ownership and rooms; the ownership price is a goal-set field (2.2 nats in shared weeks, 7.8 in own-role units).

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

H94 asks whether the agent $\times$ repo allocation of DQ4 work is a maximum-entropy (Foley) equilibrium given its constraints. The data are 24 testable units in #31–#51, in 30-min work quanta grouped into episodes. A hierarchy of max-ent models adds agent activity, repo size, ownership and rooms. After these constraints, the information left above the run-persistence floor is ≤ 0.13 of $I(\text{agent}; \text{repo})$ in 23 of 24 units, and inside the floor's 95% band in 12. Ownership is a price set by the goal: median 2.2 nats in shared weeks, 7.8 in own-role units ($p = 0.0007$). Agents do not specialize beyond ownership. Episode concentration is at or above neutral copying in shared weeks; kickoff-named repos explain it in 2 of 6. The holdout is not run.



Observables. Bits of agent–repo information per work quantum, D_1 , split by what explains it: the ownership constraint, the room constraint, the run-persistence floor (what a max-ent draw with each agent-day's runs kept would show) and the residual. Shared-goal units on the left, own-artifact units on the right. Ownership removes 79–100% of 2.7–4.1 bits in own-role units. In #31a and #33, plain max-ent with margins already sits at the floor.

PHYSICAL MODEL

Models 10 and 05: a max-ent model of a categorical allocation, with repos as replicator abundances. A work quantum is one agent $\times$ 30-min window $\times$ repo with an agent commit; an episode is a run of quanta on one repo. The owner of a repo is its first committer in DQ4. λ_{own} and λ_{room} are Lagrange multipliers (prices, in nats). D_k is the KL divergence from the level- k max-ent table. κ places the repo-size law between Maxwell–Boltzmann (0, even) and Bose–Einstein (1, neutral Pólya copying).

$$P_k(i, j) \propto a_i b_j e^{\lambda_{\text{own}} \text{own}_{ij} + \lambda_{\text{room}} \text{room}_{ij}}$$

$$D_k = \text{KL}(\hat{P} \| P_k), \quad D_1 = I(\text{agent}; \text{repo}), \quad D_{k-1} - D_k = \text{KL}(P_k \| P_{k-1})$$

$$\kappa = \frac{H_{\text{MB}} - H_{\text{Obs}}}{H_{\text{MB}} - H_{\text{BE}}}$$

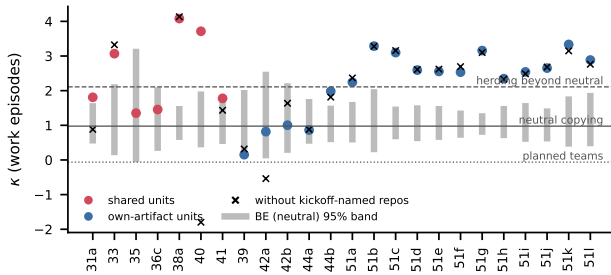
SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Work allocation in these swarms is statistically featureless once you know who works how much, which repos are big, who owns what and which room each agent sits in. There is no hidden plan to recover and no extra specialization. A monitor can therefore summarize allocation with two numbers per week: the ownership price λ_{own} and the concentration κ . The price is a lever: own-artifact goals raise it from about 2 to 8 nats, and a merge into one shared target (NE42) lowers it by ≥ 7 nats. To spread work, assign owners; to pool work, name one shared repo in the kickoff. Herding beyond neutral copying appears in 5 of 7 shared weeks, but kickoff naming accounts for it only twice.

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RESULTS VS PREDICTIONS

Prediction	Observed	Verdict
P1 $D_1 - \text{floor} \geq 0.5$ bit everywhere	20/24; #31a 0.04, #33 -0.07	failed
P2 ownership ≥ 0.5 own, < 0.5 shared	17/17 own; 6/7 shared	supported
P3 residual $\leq 0.2 D_1$ (HH285)	23/24; in 95% band 12/24	supported
P4 $\lambda_{\text{own}} \text{ own} > \text{shared}$	7.8 vs 2.2 nats, $p = 0.0007$	supported
P5 $\kappa \geq 1$ shared, < 1 own	7/7 shared; 4/17 own	mixed
P6 named repos carry κ	2/6	failed
P7 singletons from ownership	better than M1 24/24; $\pm 25\%$ 9/24	failed
P8 specialization beyond ownership	breadth < 0.8 in 1/24	failed
N G40 NE42 merge	$\lambda_{\text{own}} -7$ nats, $\kappa 0.16 \rightarrow 3.6$	supported
N G44 planned vs free arm	planned arm owns more (10.2 vs 4.1)	failed



Second figure. Concentration κ of work episodes per repo (dots) against the 95% band of Bose–Einstein (neutral copying) draws at the same K and N (gray). Crosses: without kickoff-named repos. Lines: synthetic references for neutral copying, herding beyond neutral and planned teams.

GOAL PERIODS IN DETAIL

G30 d descriptive
 G31 $\sim$ mixed
 G33 $\sim$ mixed
 G35 $\sim$ mixed
 G36 $\sim$ mixed
 G37 d descriptive
 G38 $\sim$ mixed
 G39 $\checkmark$ supported
 G40 $\checkmark$ supported
 G41 $\checkmark$ supported

+3 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	2
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	0
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

The persistence floor, the episode-level κ and the breadth test were adopted after the synthetic checks and before the real run (A1); the M3 solver and the agent-level κ are post hoc (A2). Max-ent with row, column and ownership constraints is close to saturated when most repos have one committer, so small residuals are partly by construction. In own-role units κ tracks agent activity, not herding. Owners before DQ4 and private repos are unseen. 12 units with < 100 quanta are descriptive.

ROUND 2

A reach (agents-per-repo) max-ent for herding free of activity; DQ6 rival pairs as a fourth constraint for #51; λ_{own} by day vs kickoff wording.

- **H94-R1** A herding index free of activity heterogeneity.
- **H94-R2** Close the #51 residual.
- **H94-R3** The ownership price as a phase-diagram axis.

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